Get Ready For The Quantum Computers Coming This Year

At least three companies are expected to introduce their quantum computers in 2020, while some more may be working secretly on them. In India, the Tata Institute of Fundamental Research has undertaken to lead this national mission

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Fundamentally, quantum computing is difficult to achieve, and there are some commercially viable applications for a technology that are still in their infancy. A quantum computer uses quantum bits or qubits, instead of the classical bits, which are the binary numbers, 0 and 1.

A qubit is a two-state or two-level quantum mechanical system. While a classical bit either has a value 0 or 1, a qubit can be both 0 and 1. That is, it can achieve both the values simultaneously, and the laws of quantum physics ensure that such a state is possible.

As a result, theoretical physicists estimate that a quantum computer with only about one hundred of these qubits could in principle exceed the computing power of the current classical computers. Building a quantum computer is therefore one of the main technological goals in modern physics and astrophysics. Scientists from all over the world have found a way to manufacture stable qubits that operate at room temperature, in contrast to the majority of existing analogues. This opens up new prospects for creating a quantum computer.

In the current scenario, an unprecedented data-crunching effort that aims to enhance multi-fold computing power to accelerate research for Covid-19 treatment would be most welcome. Some of the applications for solving highly complex problems, which cannot be solved by the present-day computers, are new drug discovery, supply chain logistics, financial services, artificial intelligence (AI) and cloud security.

Chemistry helps make up our world—yet there is still a lot we don't know because most of our powerful classical computers are limited in the chemical modelling they can perform and the solutions they can unlock. But quantum computing could change that completely, thus enabling large scale chemical modelling so as to unlock solutions.